

Supplementary Disease Guide

Potato Early Blight & Bell Pepper Bacterial Spot — companion supplement to the Pakistan Vegetable & Field Crop Cultivation Guide

Potato — Early Blight

Alternaria solani

Early blight is a common fungal disease of potato in Pakistan, distinct from late blight, and is caused by the fungus *Alternaria solani*. It typically appears earlier in the season than late blight and progresses more slowly, but can still cause significant yield loss if left unmanaged, especially in fields under nutrient or water stress.

Symptoms

Early blight first appears as small, dark brown to black spots on older, lower leaves, which enlarge over time into circular lesions with a distinctive “target-board” or concentric ring pattern. As the disease progresses, affected leaves yellow, dry out, and drop prematurely, reducing the plant's photosynthetic capacity and, in turn, tuber size and yield. Dark, sunken lesions can also form on tubers, particularly around the eyes, affecting storage quality.

Favorable Conditions

The disease develops fastest under warm temperatures (24–29°C) alternating with periods of high humidity or leaf wetness, and is more severe in plants weakened by drought stress, poor nutrition (especially nitrogen deficiency), or heavy insect damage.

Treatment and Management

- Use certified, disease-free seed tubers and avoid saving seed from infected fields.
- Apply balanced fertilization, particularly adequate nitrogen and potassium, since stressed or nutrient-deficient plants are far more susceptible.
- Practice crop rotation of at least two to three years away from potato, tomato, and other solanaceous crops to reduce fungal spore buildup in the soil.
- Remove and destroy infected plant debris after harvest rather than leaving it in the field.
- Apply protectant fungicides (such as those containing chlorothalonil or mancozeb) at the first sign of disease, and continue on a schedule during favorable weather conditions.
- Ensure adequate plant spacing and avoid overhead irrigation where possible, since good airflow and drier foliage reduce disease spread.

Bell Pepper (Capsicum) — Bacterial Spot

Xanthomonas campestris pv. *vesicatoria*

Bacterial spot is one of the most damaging bacterial diseases of bell pepper in warm, humid growing regions of Pakistan, caused by *Xanthomonas* species. Unlike the fungal and viral diseases mentioned elsewhere in

the pepper disease section, bacterial spot spreads rapidly through splashing water and contaminated tools, and cannot be controlled with standard fungicides.

Symptoms

Small, water-soaked spots appear on leaves, stems, and fruit, which enlarge into brown, scabby, raised lesions, often with a yellow halo on leaves. Severely affected leaves yellow and drop, weakening the plant and exposing fruit to sunscald, while fruit lesions are dark, raised, and scab-like, making produce unmarketable even when the fruit is otherwise usable.

Favorable Conditions

The disease spreads fastest in warm (24–30°C), humid or rainy conditions, and is easily transmitted by splashing irrigation water, wind-driven rain, contaminated hands and tools, and infected seed or transplants.

Treatment and Management

- Use certified, pathogen-free seed and transplants, since the bacterium can be seed-borne.
- Avoid overhead irrigation; use drip irrigation instead to keep foliage dry and reduce splash-driven spread.
- Practice crop rotation of at least one to two years away from peppers, tomatoes, and other solanaceous hosts.
- Remove and destroy infected plant debris promptly, and avoid working in fields when foliage is wet to prevent mechanical spread on hands, clothing, and tools.
- Apply copper-based bactericides (often combined with mancozeb) preventively before disease onset in high-risk weather, since bactericide sprays are far more effective as prevention than as a cure once infection is established.
- Select bacterial-spot-tolerant or resistant varieties where available from local seed suppliers, as tolerance varies significantly between hybrids.